

Loan Approval Prediction using Machine Learning

InternSpark Data Science Internship - Task 2

Objective

Build a supervised ML model to predict loan approval using borrower information.

Data Preprocessing

Missing values handled.

Label Encoding applied.

Feature Scaling using StandardScaler.

SMOTE used for class balancing.

Machine Learning Models

Logistic Regression

Random Forest Classifier

Model Performance

Metric	Logistic Regression	Random Forest
Accuracy	83.74%	80.49%
Precision	84.95%	84.27%
Recall	92.94%	88.24%
F1 Score	88.76%	86.21%
ROC-AUC	0.833	0.772

Business Interpretation

Applicants with good credit history are more likely to receive loan approval. Logistic Regression achieved the best overall performance. A deployment threshold around 0.60 is recommended. Retrain the model periodically as new data becomes available.

Conclusion

Logistic Regression outperformed Random Forest and is recommended for deployment due to its higher Accuracy, Recall, F1-score and ROC-AUC.